

QORLA Stability Test 1

Operator Conditioning, Forward-Preserving Gradient Stabilization, and Threshold Selection

QORLA Project — Experimental Research Note

Abstract

This document records the first stability optimization campaign for QORLA. The objective was to stabilize the local-linear correction while preserving the forward operator and its extrapolative behavior. The study progressed through fixed ridge, variance-adaptive ridge, mass-adaptive ridge, centered moments, a hard conditioning floor, and finally condition-aware backward damping. Forward-changing conditioning rules improved some numerical metrics but degraded difficult low-mass regression regimes. The retained mechanism therefore leaves the forward output unchanged and dampens only the backward contribution of ill-conditioned local-linear corrections. A threshold sweep over $\tau_g \in \{0.01, 0.02, 0.03, 0.05, 0.08, 0.10\}$ selected $\tau_g = 0.03$ under explicit gradient-fidelity constraints. Multi-seed confirmation showed substantial reduction of gradient spikes with no observed numerical failures and no change in the normal-case mean final loss for the selected threshold.

1 Purpose and scope

The stability campaign answers a narrow question: *can the local-linear component of QORLA be made safer to train without changing the forward attention operator?* The experiments deliberately isolate numerical conditioning, coefficient behavior, affine reproduction, gradient magnitude, gradient direction, and multi-step optimization. They do not yet constitute end-to-end model validation.

The working sequence length for the main operator-level study was $N = 256$, with Laplace scale $\lambda = 4$, baseline ridge $\rho_0 = 10^{-3}$, and $\epsilon = 10^{-8}$. Multi-seed optimization used $N = 96$, five random seeds, Adam with learning rate 3×10^{-3} , and 100 optimization steps.

2 Current QORLA forward operator

For projected query coordinate a_i and key coordinate b_j , define

$$z_{ij} = b_j - a_i, \quad W_{ij} = \exp(-\lambda|a_i - b_j|) \quad (1)$$

for the single-scale stability experiments. The multi-scale extension may replace W_{ij} by a query-conditioned positive mixture without changing the stability logic developed here.

The weighted raw moments are

$$S_{0,i} = \sum_j W_{ij}, \quad S_{1,i} = \sum_j W_{ij} z_{ij}, \quad S_{2,i} = \sum_j W_{ij} z_{ij}^2, \quad (2)$$

$$T_{0,i} = \sum_j W_{ij} v_j. \quad (3)$$

Define the weighted local displacement mean

$$\mu_i = \frac{S_{1,i}}{S_{0,i} + \epsilon} \quad (4)$$

and centered sufficient statistics

$$Q_i = \sum_j W_{ij} (z_{ij} - \mu_i)^2, \quad (5)$$

$$C_i = \sum_j W_{ij} (z_{ij} - \mu_i) v_j. \quad (6)$$

The output is decomposed into a kernel-average base term and a local-linear correction:

$$B_i = \frac{T_{0,i}}{S_{0,i} + \epsilon}, \quad (7)$$

$$R_i = -\mu_i \frac{C_i}{Q_i + \rho + \epsilon}, \quad (8)$$

$$o_i = \boxed{B_i + R_i}. \quad (9)$$

The centered form is algebraically equivalent, up to finite-precision placement of ϵ , to the raw-moment local-linear estimator. Its main role is cleaner numerical bookkeeping and a transparent decomposition into stable averaging plus a corrective regression term.

3 Stability metrics

The experiments track the following quantities.

- **Denominator margin:** $D_i = S_{0,i}(S_{2,i} + \rho) - S_{1,i}^2 + \epsilon$.
- **Condition score:**

$$\kappa_i = \frac{Q_i + \rho}{S_{2,i} + \rho + \epsilon}. \quad (10)$$

Smaller κ_i indicates a more fragile local-linear geometry.

- **Coefficient normalization error:** $\max_i |\sum_j \alpha_{ij} - 1|$.
- **Maximum effective coefficient:** $\max_{ij} |\alpha_{ij}|$.
- **Gradient magnitudes:** $\|\nabla_a\|_2$, $\|\nabla_b\|_2$, and $\|\nabla_v\|_2$.
- **Gradient fidelity:** cosine similarity between the baseline and stabilized gradients.
- **Affine reproduction:** error on $v(b) = 0.7 + 1.3b$.
- **Optimization robustness:** final loss, maximum gradient, mean gradient, and failure count over multiple seeds.

4 Experimental progression

4.1 Fixed ridge baseline

The baseline uses $\rho = \rho_0 = 10^{-3}$. It preserves affine behavior well but exhibits large key-coordinate gradients in clustered and low-kernel-mass geometries.

4.2 Variance- and mass-adaptive ridge

Two forward-changing alternatives were examined:

$$\rho_i^{\text{var}} = \rho_0 + \beta \text{Var}_i, \quad (11)$$

$$\rho_i^{\text{mass}} = \rho_0 + \beta \text{Var}_i + \frac{\gamma}{S_{0,i} + \epsilon}. \quad (12)$$

An early parameterization that allowed the ridge to fall toward 10^{-4} produced severe clustered-gradient amplification, with $\|\nabla_b\|_2$ increasing from approximately 6.14 under fixed ridge to about 121.22 for variance adaptation and 77.70 for mass adaptation. The revised forms anchored at ρ_0 were less unstable, but both increased affine-reproduction error from approximately 8.7×10^{-5} to about 6.3×10^{-4} . They were therefore rejected as default forward rules.

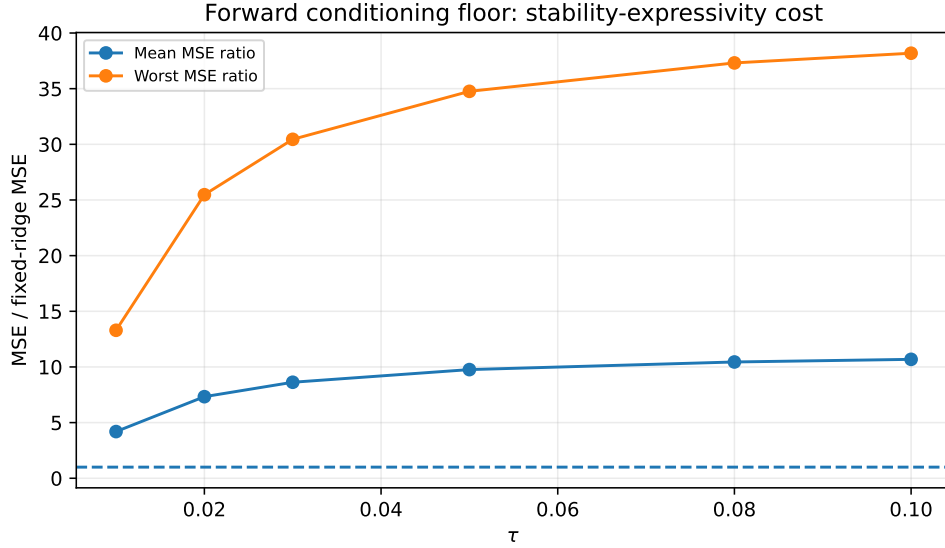


Figure 1: Forward conditioning floors reduce instability at the cost of large regression-error increases in difficult extrapolation regimes.

4.3 Centered formulation

The centered representation was tested against the raw-moment form. The output relative error remained near floating-point tolerance in ordinary cases and of order 10^{-5} in clustered/low-mass stress cases. Gradient norms remained essentially unchanged, showing that centering is a numerical representation improvement rather than a solution to the underlying conditioning problem. It is retained as the preferred algebraic form.

4.4 Hard conditioning floor

A direct conditioning floor was tested by increasing the ridge only when $\kappa_i < \tau$. This strongly reduced large coefficients and gradient spikes, but it altered the forward function. In difficult low-mass extrapolation, the price was unacceptable: even $\tau = 0.01$ increased the worst regression MSE by about $13.29\times$, while $\tau = 0.05$ increased it by about $34.76\times$. The hard floor was therefore rejected despite its strong numerical stabilization.

4.5 Condition-aware backward damping

The successful strategy preserves the forward value and modifies only the backward contribution of the correction term. Define

$$s_i = \min\left(1, \frac{\text{sg}(\kappa_i)}{\tau_g}\right) \quad (13)$$

and use

$$\tilde{R}_i = \text{sg}(R_i) + s_i (R_i - \text{sg}(R_i)). \quad (14)$$

Numerically, $\tilde{R}_i = R_i$ in the forward pass. During backpropagation, the local-linear correction gradient is multiplied by s_i . Hence

$$\boxed{o_i = B_i + R_i \quad \text{in the forward pass}} \quad (15)$$

while poorly conditioned correction gradients are selectively damped.

5 Operator-level stress tests

Five geometries were used: normal random coordinates, clustered coordinates, boundary queries, low kernel mass, and near ties. Figure 2 shows the key-coordinate gradient magnitude for the main stabilization variants.

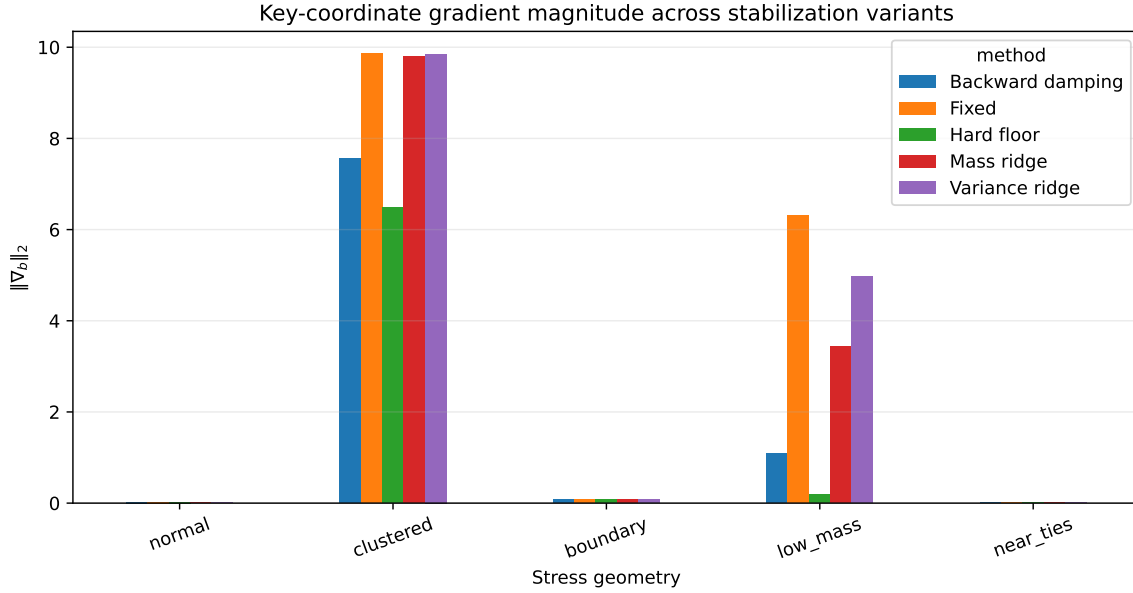


Figure 2: Key-coordinate gradient magnitude across stress geometries and stabilization variants at the operator level.

The stress results reveal two principal danger zones: moderately clustered geometry and low-kernel-mass geometry. Near ties and ordinary geometry were comparatively well behaved. The hard floor sharply suppresses gradients but changes the forward output, whereas backward damping leaves the forward operator unchanged.

6 Affine reproduction

Affine reproduction is a key guardrail because local-linear correction should preserve first-order behavior. The fixed, hard-floor-at-0.03, and backward-damping forms produced the same affine MAE of approximately 8.7×10^{-5} in the healthy test where the hard floor did not activate. Variance and mass ridge variants increased the error by roughly one order of magnitude.

7 Selecting the damping threshold

The candidate thresholds were

$$\tau_g \in \{0.01, 0.02, 0.03, 0.05, 0.08, 0.10\}. \quad (16)$$

The selection procedure was explicit rather than heuristic. A candidate was declared *admissible* only when all of the following worst-case gradient-direction constraints held:

$$\min \cos(g_a, \tilde{g}_a) \geq 0.90, \quad (17)$$

$$\min \cos(g_b, \tilde{g}_b) \geq 0.95, \quad (18)$$

$$\min \cos(g_v, \tilde{g}_v) \geq 0.95. \quad (19)$$

Among admissible thresholds, the selected value was the one with the smallest mean key-gradient norm ratio.

The sweep yielded admissible values 0.01, 0.02, and 0.03. The next candidate, 0.05, failed the fidelity constraints because the worst query-gradient cosine dropped below 0.90 and the worst key/value gradient cosines dropped below the required bounds. Among the admissible values, $\tau_g = 0.03$ provided the strongest damping.

At $\tau_g = 0.03$, the sweep gave mean key-gradient cosine similarity ≈ 0.9948 , worst key-gradient cosine ≈ 0.9799 , worst query-gradient cosine ≈ 0.9232 , worst value-gradient cosine ≈ 0.9587 , and mean key-gradient norm ratio ≈ 0.407 . Thus the average key-gradient magnitude was reduced by approximately 59% while preserving direction closely.

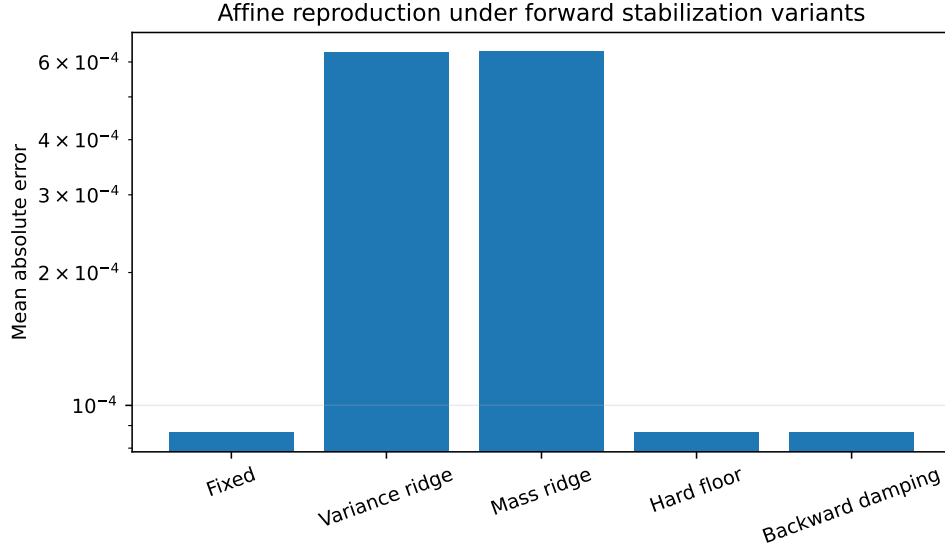


Figure 3: Affine reproduction error for forward stabilization variants. Backward damping leaves the forward operator unchanged and therefore preserves the baseline error.

8 Multi-step and multi-seed confirmation

Single-step fidelity is not sufficient; the modified gradients must also support actual optimization. The final confirmation used three geometry classes (normal, cluster, low mass), five seeds, 100 Adam steps, and compared fixed training with $\tau_g = 0.03$ and $\tau_g = 0.05$.

The selected threshold left the normal-case mean final loss unchanged to the displayed precision. In the low-mass case, it reduced the mean maximum key-gradient from about 182.78 to 45.64 and the worst observed spike from 224.30 to 58.02, with zero failures. The stronger 0.05 threshold damped gradients further but showed greater interference with ordinary optimization and had already failed the single-step fidelity constraints.

9 Final retained stability configuration

The current stability configuration is

$$\kappa_i = \frac{Q_i + \rho}{S_{2,i} + \rho + \epsilon}, \quad s_i = \min \left(1, \frac{\text{sg}(\kappa_i)}{0.03} \right) \quad (20)$$

with

$$\tilde{R}_i = \text{sg}(R_i) + s_i (R_i - \text{sg}(R_i)). \quad (21)$$

The forward operator is unchanged. The selected rule only modifies the gradient contribution of the local-linear correction when local conditioning falls below the selected threshold.

The main conclusion is therefore not that large ridge values make QORLAs stable. Instead, the experiments support a more targeted principle:

Preserve the expressive forward local-linear operator, diagnose fragile geometry through a conditioning score, and damp only the unstable backward correction.

10 Limitations and next validation

This first campaign is complete at the synthetic/operator level, but the selected threshold remains provisional. The next required validation is inside a trainable attention model in which query/key projections, values, and upstream

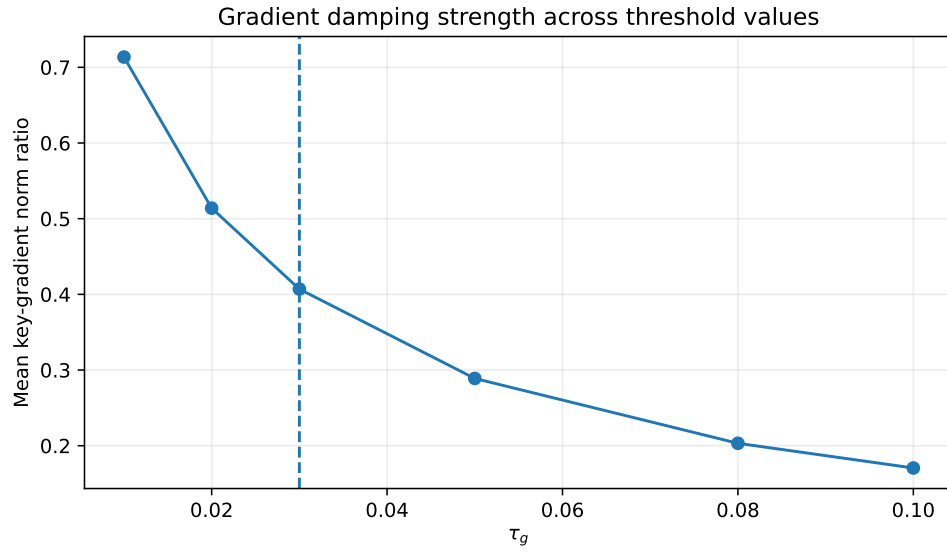


Figure 4: Mean key-gradient norm ratio versus damping threshold. Larger thresholds provide stronger damping.

representations evolve jointly. The present study establishes a reproducible default for that stage, not a universal theorem that 0.03 is optimal for every architecture, scale bank, precision mode, or dataset.

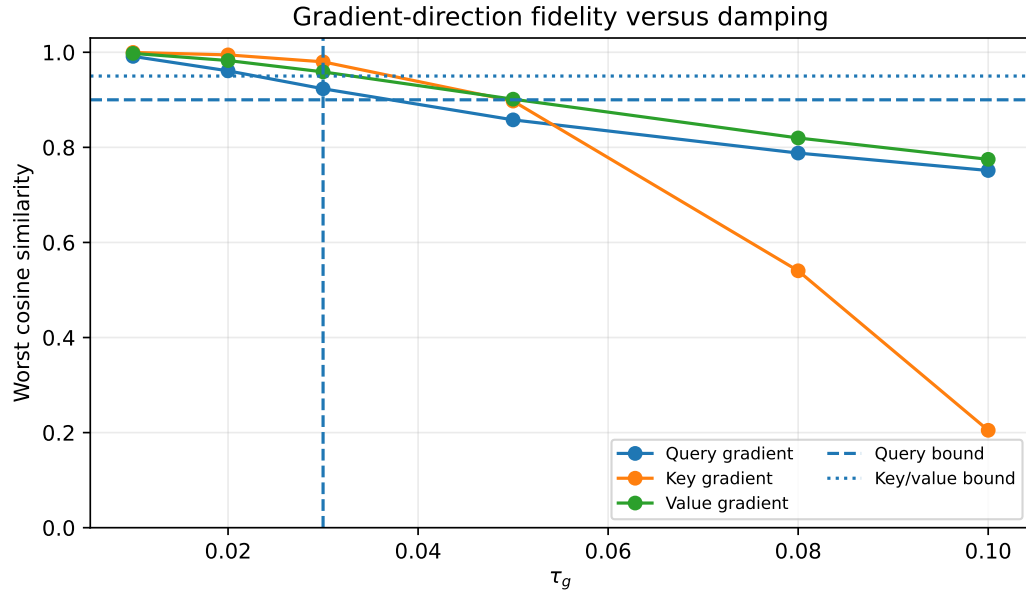


Figure 5: Worst gradient-direction cosine similarities across thresholds. The selected $\tau_g = 0.03$ remains inside the predefined fidelity region; $\tau_g = 0.05$ does not.

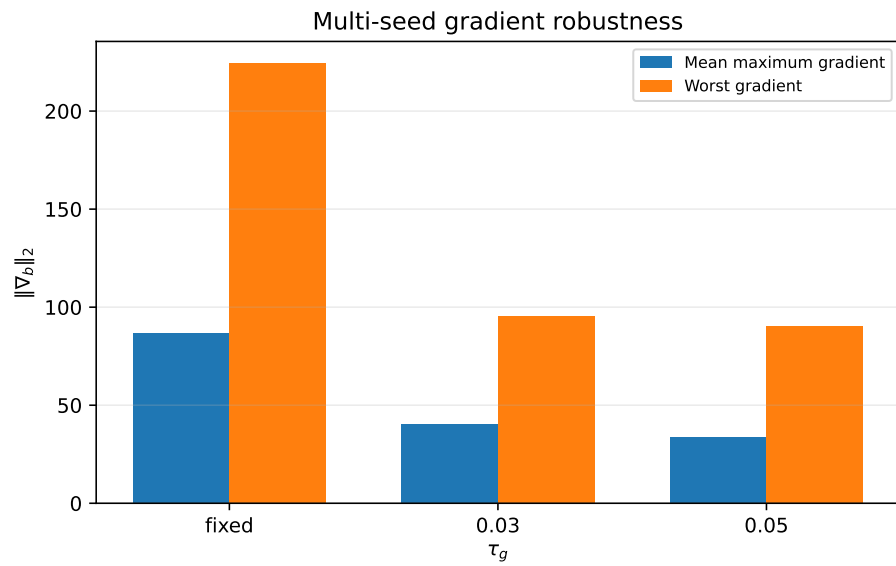


Figure 6: Multi-seed comparison of average maximum and worst key-coordinate gradient magnitude.

A Full operator stress table

Case	Method	Loss	$\min D$	$\min \kappa$	$\max \alpha $	$\ \nabla_b\ _2$	Sum error
normal	Fixed	0.04402	0.009695	0.1963	1.078	0.02362	7.75e-07
normal	Variance ridge	0.04064	0.01357	0.2548	0.9012	0.01931	4.77e-07
normal	Mass ridge	0.04022	0.01457	0.2685	0.8708	0.01909	4.77e-07
normal	Hard floor	0.04402	0.009695	0.1963	1.078	0.02362	7.75e-07
normal	Backward damping	0.04402	0.009695	0.1963	1.078	0.02362	7.75e-07
clustered	Fixed	0.01938	0.2876	0.008374	0.06834	9.855	1.56e-05
clustered	Variance ridge	0.01938	0.2876	0.008375	0.06833	9.853	1.38e-05
clustered	Mass ridge	0.01927	0.2886	0.008404	0.06811	9.789	1.64e-05
clustered	Hard floor	0.01379	0.3055	0.03	0.03746	6.477	5.25e-06
clustered	Backward damping	0.01938	0.2876	0.008374	0.06834	7.565	1.56e-05
boundary	Fixed	0.02554	35.11	0.1914	0.05164	0.08074	8.34e-07
boundary	Variance ridge	0.02548	35.24	0.1919	0.05151	0.08049	7.15e-07
boundary	Mass ridge	0.02547	35.24	0.192	0.05151	0.08048	8.34e-07
boundary	Hard floor	0.02554	35.11	0.1914	0.05164	0.08074	8.34e-07
boundary	Backward damping	0.02554	35.11	0.1914	0.05164	0.08074	8.34e-07
low_mass	Fixed	0.5987	0.0005644	0.00416	0.3924	6.304	5.17e-05
low_mass	Variance ridge	0.4714	0.0007367	0.005006	0.3542	4.964	4.97e-05
low_mass	Mass ridge	0.325	0.001737	0.005909	0.3341	3.437	4.05e-05
low_mass	Hard floor	0.01761	0.004127	0.03	0.08163	0.19	7.51e-06
low_mass	Backward damping	0.5987	0.0005644	0.00416	0.3924	1.087	5.17e-05
near_ties	Fixed	0.01077	64.13	0.504	0.0613	0.008911	2.38e-07
near_ties	Variance ridge	0.01076	64.33	0.5048	0.06121	0.008896	2.38e-07
near_ties	Mass ridge	0.01076	64.33	0.5048	0.06121	0.008896	2.38e-07
near_ties	Hard floor	0.01077	64.13	0.504	0.0613	0.008911	2.38e-07
near_ties	Backward damping	0.01077	64.13	0.504	0.0613	0.008911	2.38e-07

B Affine reproduction table

Method	MAE	Max error	$\min \kappa$
Fixed	8.668e-05	6.501e-04	0.2359
Variance ridge	6.297e-04	4.596e-03	0.2371
Mass ridge	6.335e-04	4.640e-03	0.2371
Hard floor	8.668e-05	6.501e-04	0.2359
Backward damping	8.668e-05	6.501e-04	0.2359

Table 2: Affine reproduction under the tested forward stabilization variants.

C Threshold-selection table

D Multi-seed confirmation table

E Experiment chronology

τ_g	Mean $\cos_a$	Min $\cos_a$	Mean $\cos_b$	Min $\cos_b$	Mean $\cos_v$	Min $\cos_v$	Mean grad_b ratio	Pass
0.01	0.9932	0.9913	0.9999	0.9997	0.9989	0.9976	0.7135	Yes
0.02	0.9779	0.9609	0.9986	0.9945	0.9918	0.9826	0.5139	Yes
0.03	0.9600	0.9232	0.9948	0.9799	0.9805	0.9587	0.4070	Yes
0.05	0.9297	0.8578	0.9733	0.8973	0.9527	0.9013	0.2889	No
0.08	0.8966	0.7880	0.8749	0.5403	0.9126	0.8197	0.2032	No
0.10	0.8802	0.7513	0.7320	0.2049	0.8900	0.7748	0.1706	No

Table 3: Gradient-fidelity sweep used to select the damping threshold.

Case	τ_g	Mean final loss	Mean max grad_b	Worst grad_b	Mean grad_b	Failures
normal	fixed	12.1642	3.079	6.562	2.068	0
normal	0.03	12.1642	3.079	6.562	2.068	0
normal	0.05	12.1712	3.018	6.562	2.019	0
cluster	fixed	5.66285e-05	73.69	96.43	2.302	0
cluster	0.03	6.36164e-05	72.69	95.38	2.268	0
cluster	0.05	3.89388e-05	68.08	90.15	2.188	0
lowmass	fixed	0.330129	182.8	224.3	30.27	0
lowmass	0.03	0.337899	45.64	58.02	7.806	0
lowmass	0.05	0.342259	29.68	37.64	5.144	0

Table 4: Five-seed, 100-step optimization confirmation.

Stage	Mechanism	Decision
STAB-01	Fixed, variance, and mass ridge	Naive adaptive ridge rejected; early low-ridge parameterization amplified clustered gradients.
STAB-02	Centered moments	Retained as the preferred numerical representation; did not materially alter the underlying conditioning.
STAB-03	Hard conditioning floor	Numerically effective but forward-changing.
STAB-04	Healthy regression geometries	Conditioning floor remained silent when κ was already large.
STAB-05	Ill-conditioned regression	Hard floor rejected because low-mass extrapolation error increased sharply.
STAB-06	Backward-only damping	Passed: forward difference was zero while gradients were substantially reduced.
STAB-07	Gradient fidelity sweep	Established admissibility constraints and identified 0.03 as the strongest admissible threshold.
STAB-08	Multi-step optimization	Confirmed that damping can improve or preserve optimization while reducing spikes.
STAB-09	Multi-seed confirmation	Supported 0.03 as the conservative default for subsequent experiments.

Table 5: Chronology of the first QORLA stability campaign.